

Senthil Vel Kumaran

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📍 Chennai, India

PROFESSIONAL SUMMARY

Automotive Embedded Systems Engineer with 2+ years delivering safety-critical firmware for cockpit ECUs deployed in **400,000+ Mahindra and Tata Motors production vehicles** via Visteon Corporation (Tier-1 OEM supplier). Sole module owner of DCM, DEM, AUTOSAR CAN stack, MDIO driver, and Ethernet switch configuration across **3 shipped ECU programs** authoring 10,000+ lines of production Embedded C. Experienced across AUTOSAR Classic, ISO 14229 (UDS), ISO 26262 (ASIL-D), and real-time in-vehicle protocols (CAN FD, Automotive Ethernet, SPI, I²C, UART), with hands-on interfacing to Qualcomm Snapdragon 8155 SoC and QNX Neutrino RTOS.

EXPERIENCE

Visteon Corporation (Tier-1 Automotive Supplier Mahindra & Tata Motors OEM Programs) Jul. 2023 – Present
Embedded Systems Engineer Chennai, India

- Firmware engineer on 3 production ECU programs for Mahindra and Tata Motors cockpit platforms; **software runs in 400,000+ production vehicles** sole contributor of 10,000+ lines of Embedded C across all programs
- Sole owner of DCM and DEM modules across 2 programs; implemented **13 UDS service handlers** (ISO 14229) covering session control, security access, DTC read/clear, routine control, and data I/O delivered complete diagnostic stack ahead of program gate, earning the Visteon Silver Award
- Owned AUTOSAR CAN communication stack end-to-end: authored DBC/ARXML-based signal configuration for **200+ PDUs**, implemented MCAL CAN driver, CanIf, PduR, and COM layer handlers for Classic CAN and CAN FD on 2 production ECUs
- Wrote Clause 22 MDIO driver from scratch for Realtek Ethernet PHY covering register-map initialization, auto-negotiation sequencing, and link-status polling and configured NXP SJA1105 Ethernet switch via SPI; completed 100BASE-T1 bring-up **ahead of hardware integration milestone**
- Designed custom UART framing protocol for IPC between Renesas RH850 MCU and Qualcomm Snapdragon 8155 SoC; implemented message packetization, sequence numbering, and error recovery to deliver live vehicle CAN signals into QNX Neutrino RTOS and Android Automotive HAL
- Implemented ISO 26262 ASIL-D thermal safety monitor on NXP MPC560xP: periodic I²C ADC sampling, multi-threshold hysteresis logic, and PWM-based backlight attenuation; **zero thermal-related field escapes** across full OEM validation cycle
- Reduced ECU quiescent current by **20–40%** on Renesas RH850 through wake-up source filtering, peripheral clock gating, and AUTOSAR EcuM sleep sequence optimization
- Resolved **500+ MISRA-C and static analysis defects** (KlocWork, Coverity) across 3 production codebases; achieved zero high-severity escape at integration review gate

Visteon Corporation Jan. 2023 – Jun. 2023
Embedded Software Engineer Intern Chennai, India

- Led display subsystem bring-up for triple 12.5-inch cockpit on Infineon Traveo II: configured Maxim GMSL2 serializer/deserializer chain, I²C-based deserializer register map initialization, pixel clock synchronization, and forward channel link training **completed ahead of schedule**, unblocking downstream software integration
- Developed and validated SPI and UART peripheral drivers at register level; traced execution flow and inspected memory-mapped peripheral registers using Lauterbach Trace32; decoded and verified live CAN frame timing with Vector CANalyzer and CANoe
- Audited 3 production codebases using KlocWork and Coverity; **resolved 500+ MISRA-C and static analysis violations** spanning null-pointer dereference, uninitialized variables, and implicit type conversion cleared entire defect backlog before first hardware-in-loop test

PROJECTS

Triple 12.5-inch Cockpit HPC Controller Tata Motors | Infineon Traveo II (ARM Cortex-M7), AUTOSAR Classic, 100BASE-T1

- Configured AUTOSAR OS (tasks, alarms, category-2 ISRs) and integrated full CAN communication stack (MCAL → CanIf → PduR → COM) with DBC-based ARXML signal mapping for a production cockpit ECU driving triple simultaneous displays
- Sole owner of Automotive Ethernet bring-up: wrote Clause 22 MDIO driver for Realtek PHY initialization, link-up detection, and speed negotiation; configured NXP SJA1105 L2 switch forwarding tables and VLAN assignments via SPI over 100BASE-T1

Dual 10.25-inch Cockpit Domain Controller Mahindra | Renesas RH850, Qualcomm Snapdragon 8155, QNX Neutrino

- Designed custom UART IPC layer between Renesas RH850 MCU and Qualcomm Snapdragon 8155 SoC: byte-level framing, CRC validation, and sequence-number-based retransmission; vehicle CAN signals delivered to QNX Neutrino and Android Automotive HAL with sub-10ms latency
- Sole owner of UDS diagnostic stack (DCM + DEM): 13 ISO 14229 service handlers, DTC fault memory with freeze frame capture, event-based DTC aging, and OBD-II readiness flag management validated against Mahindra diagnostic test specification

10.25-inch Standalone Infotainment ECU (Fuel + EV) | NXP MPC560xP, ISO 26262 ASIL-D, I²C, SerDesTeam of 5

- Implemented ASIL-D thermal safety monitor: periodic I²C ADC reads from NXP temperature sensor IC, multi-level hysteresis thresholds (warning / derate / shutdown), and PWM backlight attenuation runtime safety state reported via DEM to the host SoC
- Performed Maxim MAX96705/MAX9272 SerDes bring-up for both fuel and EV hardware variants: I²C register initialization, COAX link training, and pixel clock lock verification; unified bring-up driver shared across both platform variants to eliminate duplicate code

EDUCATION

Vellore Institute of Technology

Jul. 2019 – Jul. 2023

B.Tech. – Electronics and Communication Engineering

Chennai, India

- **CGPA: 8.53 / 10.0** | *Relevant Coursework:* Embedded Systems, VLSI Design, Microprocessors, Real-Time OS, Digital Signal Processing

TECHNICAL SKILLS

Languages Embedded C (2+ yrs, production), C++, Python, MATLAB (basic)

Microcontrollers Renesas RH850, Infineon Traveo II (ARM Cortex-M7), NXP MPC560xP, ATmega328P (AVR)

SoC / RTOS Qualcomm Snapdragon 8155, QNX Neutrino, Android Automotive, AUTOSAR OS (OSEK)

Protocols CAN, CAN FD, Automotive Ethernet (100BASE-T1), TCP/IP, SPI, I²C, UART, MDIO, LVDS, GMSL

Frameworks AUTOSAR Classic (MCAL, CanIf, PduR, COM, DCM, DEM, NM), ISO 14229 (UDS), ISO 26262 (ASIL-D)

Display / Imaging SerDes (Maxim GMSL, LVDS), High-Speed Display Bring-Up, PWM Backlight Control, Multi-Display Pipeline

Tools Lauterbach Trace32, Vector CANalyzer, CANoe, VectorCAST, KlocWork, Coverity, Multi IDE, Git, Jira

HONOURS & AWARDS

Silver Award Delivered complete DCM & DEM diagnostic stack ahead of program schedule, enabling on-time OEM sign-off for Mahindra and Tata Motors programs; Visteon Corporation

Top Performer Ranked #1 in C Embedded Language Training cohort; Yashwant Kanetkar & Visteon Corporation